Essential Java Keywords Reference

CSCI212

Introduction

This document provides a categorized reference of essential Java keywords. Each keyword is defined conceptually and demonstrated with a short, representative line of Java code. The goal is to build both syntactic familiarity and semantic understanding.

1 Class and Object Structure Keywords

`class`

Defines a blueprint for creating objects.

```
class Student { }
```

`interface`

Defines a contract that implementing classes must follow.

```
interface Drawable { void draw(); }
```

`abstract`

Used with classes or methods that are incomplete and must be extended or implemented.

```
abstract class Shape { abstract double area(); }
```

`extends`

Allows a subclass to inherit from a superclass.

```
class Circle extends Shape { }
```

`implements`

Used by classes to implement an interface.

```
class Printer implements Drawable { }
```

2 Access Control Keywords

`public`

Makes a class, method, or variable accessible everywhere.

```
public int count;
```

`private`

Restricts access to within the same class only.

```
private double salary;
```

`protected`

Allows access within the same package or subclasses.

```
protected String name;
```

3 Object Creation and Memory Keywords

`new`

Allocates memory and creates a new object.

```
Student s = new Student();
```

`this`

Refers to the current object instance.

```
this.count = count;
```

`super`

Refers to the parent class object.

```
super.area();
```

4 Method and Variable Modifiers

`void`

Specifies that a method returns no value.

```
void printMessage() { }
```

`static`

Belongs to the class rather than instances.

```
static int totalStudents;
```

`final`

Prevents modification: final variables, methods, or classes.

```
final double PI = 3.14159;
```

`constant` (conceptual)

Java does not have a `constant` keyword; constants are declared using `static final`.

```
static final int MAX_SIZE = 100;
```

5 Exception Handling Keywords

`try`

Wraps code that may cause an exception.

```
try { int x = 10 / 0; }
```

`catch`

Handles exceptions thrown in a `try` block.

```
catch (ArithmeticException e) { }
```

`finally`

Executes regardless of whether an exception occurs.

```
finally { System.out.println("Done"); }
```

`throw`

Explicitly throws an exception.

```
throw new IllegalArgumentException();
```

`throws`

Declares exceptions that a method may pass upward.

```
void read() throws IOException { }
```

`except` (non-Java)

`except` is *not* a Java keyword; it belongs to Python. Java uses `catch` instead.

6 Control Flow Keywords

`if`

Executes code conditionally.

```
if (x > 0) { }
```

`else`

Executes when the `if` condition is false.

```
else { }
```

`for`

Creates a counting loop.

```
for (int i = 0; i < 10; i++) { }
```

`while`

Repeats while a condition remains true.

```
while (running) { }
```

`break`

Exits a loop prematurely.

```
break;
```

`continue`

Skips the current loop iteration.

```
continue;
```

7 Type and Package Keywords

`package`

Groups related classes into namespaces.

```
package edu.qc.csci212;
```

`import`

Allows access to classes defined in other packages.

```
import java.util.ArrayList;
```

`instanceof`

Tests whether an object belongs to a specific class.

```
if (obj instanceof Student) { }
```

`return`

Sends a value back from a method.

```
return total;
```